

18.1 Analysis of covariance as a projection problem

In many experiments something about each unit is known before the treatments are applied: a pretest score, the weight of an animal at the start of a feeding trial, the yield of a plot in the previous season, a patient's blood pressure at enrolment. Such a measurement is a **covariate**. If it predicts the response, much of the variation that the one-way analysis (Chapter 15) calls error is not error at all. Part of it is the covariate at work, and it can be removed.

The **analysis of covariance** (ANCOVA) adds covariates to a factor model. Section 8.6 treated one factor and one covariate (Proposition 8.6.3). This section states the result for any design part, rank deficient or not, and any number of covariates, and then asks what a covariate does in a randomized experiment.

18.1.1 The model

Write the mean of the response as the sum of a *design part* and a *covariate part*,

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n. \quad (18.1.1)$$

Here $\mathbf{X}$ is $n \times p$ with rank r , built from the indicator columns of the treatment factor and of any blocking factors. It is usually overparameterized. The $n \times q$ matrix $\mathbf{Z}$ holds the covariate values, and $\boldsymbol{\gamma} \in \mathbb{R}^q$ their coefficients. Let $\mathbf{M}$ be the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. Everything below is written with the three matrices

$$\mathbf{E}_{zz} = \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Z}, \quad \mathbf{E}_{zy} = \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{y}, \quad E_{yy} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{y}. \quad (18.1.2)$$

The letter E stands for *error* or *within*: in the one-way layout $(\mathbf{I} - \mathbf{M})\mathbf{y}$ holds the deviations from the group means (Example (The one-way layout)), so these are within-group sums of squares and products. We assume throughout that

$$\text{rank}[\mathbf{X}, \mathbf{Z}] = \text{rank}(\mathbf{X}) + q, \quad \text{equivalently} \quad (\mathbf{I} - \mathbf{M})\mathbf{Z} \text{ has full column rank } q. \quad (18.1.3)$$

The two forms are equivalent by Lemma 9.2.1, since $\mathcal{C}([\mathbf{X}, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}((\mathbf{I} - \mathbf{M})\mathbf{Z})$. The condition fails exactly when some combination of the covariates lies in the design space. With one factor and one covariate, it fails when the covariate is constant within every group. Then the covariate cannot be separated from the group effects (Exercise (B1)).

18.1.2 The main theorem

Theorem 18.1.1 — Analysis of covariance

Assume Eq. 18.1.1 and Eq. 18.1.3. Let $\mathbf{M}_W$ be the orthogonal projection onto $\mathcal{C}([\mathbf{X}, \mathbf{Z}])$.

- (a) $\mathbf{M}_W = \mathbf{M} + (\mathbf{I} - \mathbf{M})\mathbf{Z}\mathbf{E}_{zz}^{-1}\mathbf{Z}^\top(\mathbf{I} - \mathbf{M})$, and the two terms are orthogonal projections

with orthogonal ranges.

- (b) $\boldsymbol{\gamma}$ is estimable, with the unique least squares estimate $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1}\mathbf{E}_{zy}$.
 (c) A function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable in Eq. 18.1.1 iff it is estimable in the model $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ without covariates. If $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$, its least squares estimate is

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}), \quad (18.1.4)$$

so the fitted design part is $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}})$: the covariance-adjusted response $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}$ analysed by the formulas of the model without covariates.

- (d) The residual sum of squares is

$$\text{SSE} = E_{yy} - \mathbf{E}_{zy}^\top \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}, \quad (18.1.5)$$

on $n - r - q$ degrees of freedom.

- (e) $\text{Cov}(\hat{\boldsymbol{\gamma}}) = \sigma^2 \mathbf{E}_{zz}^{-1}$, $\text{Cov}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}}) = -\sigma^2 \boldsymbol{\rho}^\top \mathbf{M} \mathbf{Z} \mathbf{E}_{zz}^{-1}$, and

$$\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{\rho}^\top \mathbf{M} \boldsymbol{\rho} + \boldsymbol{\rho}^\top \mathbf{M} \mathbf{Z} \mathbf{E}_{zz}^{-1} \mathbf{Z}^\top \mathbf{M} \boldsymbol{\rho}). \quad (18.1.6)$$

- (f) Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with rank $r_0 < r$ and projection $\mathbf{M}_0$, and define $\mathbf{T}_{zz}, \mathbf{T}_{zy}, T_{yy}$ as in Eq. 18.1.2 with $\mathbf{M}_0$ in place of $\mathbf{M}$. Then Eq. 18.1.3 holds for $\mathbf{X}_0$ as well, and the hypothesis that the design part of the mean lies in $\mathcal{C}(\mathbf{X}_0)$, with $\boldsymbol{\gamma}$ unrestricted, has the **adjusted sum of squares**

$$\text{SS}_H = (T_{yy} - \mathbf{T}_{zy}^\top \mathbf{T}_{zz}^{-1} \mathbf{T}_{zy}) - (E_{yy} - \mathbf{E}_{zy}^\top \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}) \quad (18.1.7)$$

on $r - r_0$ degrees of freedom. If $\mathbf{Y}$ is normal,

$$F = \frac{\text{SS}_H / (r - r_0)}{\text{SSE} / (n - r - q)} \sim F(r - r_0, n - r - q, \gamma),$$

where the noncentrality γ , a scalar not to be confused with the coefficient vector $\boldsymbol{\gamma}$, is given by $\sigma^2 \gamma$ equal to the squared distance from $\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}$ to $\mathcal{C}([\mathbf{X}_0, \mathbf{Z}])$.

Proof.

- (a) By Lemma 9.2.1, $\mathcal{C}([\mathbf{X}, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\tilde{\mathbf{Z}})$ with $\tilde{\mathbf{Z}} = (\mathbf{I} - \mathbf{M})\mathbf{Z}$, and the summands are orthogonal. $\tilde{\mathbf{Z}}$ has full column rank by Eq. 18.1.3, so its projection is $\tilde{\mathbf{Z}}(\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}})^{-1} \tilde{\mathbf{Z}}^\top$ (Theorem 6.3.7), and $\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} = \mathbf{E}_{zz}$ because $\mathbf{I} - \mathbf{M}$ is symmetric and idempotent. Theorem 6.3.8 adds the two projections.
 (b) and (c). Put $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}$. By (a),

$$\mathbf{M}_W \mathbf{y} = \mathbf{M} \mathbf{y} + (\mathbf{I} - \mathbf{M}) \mathbf{Z} \hat{\boldsymbol{\gamma}} = \mathbf{M}(\mathbf{y} - \mathbf{Z} \hat{\boldsymbol{\gamma}}) + \mathbf{Z} \hat{\boldsymbol{\gamma}}.$$

The first term lies in $\mathcal{C}(\mathbf{X})$, so it equals $\mathbf{X}\mathbf{b}$ for some $\mathbf{b}$, and $(\mathbf{b}, \hat{\boldsymbol{\gamma}})$ is a least squares estimate (Theorem 6.4.1). If $(\mathbf{b}', \boldsymbol{\gamma}')$ is another one, then $\mathbf{X}(\mathbf{b} - \mathbf{b}') = \mathbf{Z}(\boldsymbol{\gamma}' - \hat{\boldsymbol{\gamma}})$. Applying $\mathbf{I} - \mathbf{M}$ gives $\tilde{\mathbf{Z}}(\boldsymbol{\gamma}' - \hat{\boldsymbol{\gamma}}) = \mathbf{0}$, and so $\boldsymbol{\gamma}' = \hat{\boldsymbol{\gamma}}$. The coefficient vector of the covariates is therefore the same in every least squares estimate, and so $\boldsymbol{\gamma}$ is estimable (Theorem 8.3.1(g)). The same argument shows that $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}})$ for every least squares estimate.

For estimability of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$: in the covariance model this function has coefficient vector $(\boldsymbol{\lambda}^\top, \mathbf{0}^\top)$, which is estimable iff $\mathbf{a}^\top [\mathbf{X}, \mathbf{Z}] = (\boldsymbol{\lambda}^\top, \mathbf{0}^\top)$ for some $\mathbf{a}$ (Theorem 8.2.2). Such

an $\mathbf{a}$ gives $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$. Conversely, if $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$, put

$$\mathbf{a} = \mathbf{M}\boldsymbol{\rho} - \tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{M}\boldsymbol{\rho}.$$

Then $\mathbf{a}^\top\mathbf{X} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{X} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{X} = \boldsymbol{\lambda}^\top$, because $\tilde{\mathbf{Z}}^\top\mathbf{X} = \mathbf{Z}^\top(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$, and $\mathbf{a}^\top\mathbf{Z} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{Z} = \mathbf{0}^\top$, because $\tilde{\mathbf{Z}}^\top\mathbf{Z} = \mathbf{E}_{zz}$. So $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$ is estimable. Since $\mathbf{a}$ lies in $\mathcal{C}([\mathbf{X}, \mathbf{Z}])$, its least squares estimate is $\mathbf{a}^\top\mathbf{M}_W\mathbf{y} = \mathbf{a}^\top\mathbf{y}$ (Proposition 8.3.2(a)):

$$\mathbf{a}^\top\mathbf{y} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{y} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\mathbf{E}_{zy} = \boldsymbol{\rho}^\top\mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}).$$

- (d) $\text{SSE} = \|(\mathbf{I} - \mathbf{M}_W)\mathbf{y}\|^2 = \mathbf{y}^\top(\mathbf{I} - \mathbf{M})\mathbf{y} - \mathbf{y}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{y}$ by (a), and $\tilde{\mathbf{Z}}^\top\mathbf{y} = \mathbf{E}_{zy}$. The rank of $\mathbf{M}_W$ is $r + q$.
- (e) $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{Y}$ has covariance $\sigma^2\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1} = \sigma^2\mathbf{E}_{zz}^{-1}$. The estimate of $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$ is $\mathbf{a}^\top\mathbf{Y}$, and $\mathbf{M}\boldsymbol{\rho}$ is orthogonal to $\tilde{\mathbf{Z}}$. So $\text{Var}(\mathbf{a}^\top\mathbf{Y}) = \sigma^2\|\mathbf{a}\|^2$ is the sum of the two squared lengths in Eq. 18.1.6, and $\text{Cov}(\mathbf{a}^\top\mathbf{Y}, \hat{\boldsymbol{\gamma}}) = \sigma^2\mathbf{a}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1} = -\sigma^2\boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}$.
- (f) If $(\mathbf{I} - \mathbf{M}_0)\mathbf{Z}\mathbf{c} = \mathbf{0}$, then $\mathbf{Z}\mathbf{c} \in \mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, so $(\mathbf{I} - \mathbf{M})\mathbf{Z}\mathbf{c} = \mathbf{0}$ and $\mathbf{c} = \mathbf{0}$. So Eq. 18.1.3 holds for $\mathbf{X}_0$, and (d) applied to $[\mathbf{X}_0, \mathbf{Z}]$ gives the first bracket in Eq. 18.1.7 as the residual sum of squares of the reduced model. The rank of the reduced model is $r_0 + q$, and $\mathcal{C}([\mathbf{X}_0, \mathbf{Z}]) \subseteq \mathcal{C}([\mathbf{X}, \mathbf{Z}])$. The distribution is Theorem 11.1.1.

■

The proof is the Frisch–Waugh–Lovell theorem (Theorem 6.6.2) with the covariates as the residualized columns, stated through $\mathbf{M}$ so that the design part may be rank deficient.

Idea. Analysis of covariance is two ordinary analyses joined by one projection. The within-group regression of $\mathbf{y}$ on the covariates gives $\hat{\boldsymbol{\gamma}}$, and the analysis of variance of the adjusted response $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}$ gives the treatment estimates. Their variances Eq. 18.1.6 carry an extra term for the uncertainty in $\hat{\boldsymbol{\gamma}}$. Tests must compare models, as in Eq. 18.1.7: the between-groups sum of squares of the adjusted response is not the numerator of any F statistic, because the reduced model must be allowed to re-estimate $\boldsymbol{\gamma}$.

18.1.3 The one-way layout with one covariate

For a single factor with g levels and a single covariate x , the matrices of Eq. 18.1.2 are scalars:

$$E_{xx} = \sum_k \sum_{i \in k} (x_i - \bar{x}_k)^2, \quad E_{xy} = \sum_k \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k),$$

and E_{yy} is the within-groups sum of squares of y . With $\mathbf{X}_0 = \mathbf{1}$, the T quantities are the same sums around the grand means. Theorem 18.1.1 gives:

- (i) slope $\hat{\beta} = E_{xy}/E_{xx}$;
- (ii) $\text{SSE} = E_{yy} - E_{xy}^2/E_{xx}$ on $n - g - 1$ degrees of freedom;
- (iii) adjusted treatment sum of squares $\text{SS}_{\text{trt}} = (T_{yy} - T_{xy}^2/T_{xx}) - (E_{yy} - E_{xy}^2/E_{xx})$ on $g - 1$ degrees of freedom;
- (iv) level means adjusted to a common covariate value, $\bar{y}_k - \hat{\beta}(\bar{x}_k - x_0)$, which differ by $(\bar{y}_k - \bar{y}_l) - \hat{\beta}(\bar{x}_k - \bar{x}_l)$, as in Proposition 8.6.3;
- (v) the test of $\beta = 0$, with sum of squares E_{xy}^2/E_{xx} on one degree of freedom (Corollary 9.2.3).

Example. (Four study formats with a pretest)

In a synthetic randomized experiment, $n = 40$ students are assigned at random, ten to each of four study formats for an introductory statistics unit. A pretest score x is recorded before the assignment, and the response y is the score on the final examination. The data were generated from a parallel lines model. Randomization balanced the pretest only on average: the four groups have pretest means 62.1, 57.2, 59.9 and 60.5, and final means 72.86, 75.15, 69.09 and 78.09. [Figure 18.1.1](#) shows the data, and the sums of squares and products are:

	xx	xy	yy	df
between formats	124.9	-31.4	432.94	3
within formats	3643.9	3159.2	4146.41	36
total	3768.8	3127.8	4579.35	39

Without the covariate, the one-way analysis has error mean square $4146.41/36 = 115.2$ and treatment $F = 1.25$ on 3 and 36 degrees of freedom ($p = 0.305$). There is no evidence that the formats differ.

With the covariate, the within-groups slope is $\hat{\beta} = 3159.2/3643.9 = 0.8670$ points on the final per point on the pretest. The residual sum of squares falls to

$$\text{SSE} = 4146.41 - 2738.94 = 1407.47$$

on 35 degrees of freedom, so the error mean square falls from 115.2 to 40.2. The residual standard deviation falls from 10.73 to 6.34. The reduced model, one line for all forty students, has residual sum of squares $4579.35 - 2595.83 = 1983.52$. The adjusted treatment sum of squares is therefore $1983.52 - 1407.47 = 576.05$, and

$$F = \frac{576.05/3}{1407.47/35} = 4.77$$

on 3 and 35 degrees of freedom, with $p = 0.0068$: clear differences between the formats. The test of $\beta = 0$ has $F = 68.1$. The adjusted treatment sum of squares, 576.05, is even *larger* than the unadjusted 432.94: format 2 had the weakest students on entry and still finished second, and adjustment gives it credit for that.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(84)
g, m = 4, 10                                # four formats, ten students
    each
n = g * m
group = rng.permutation(np.repeat(np.arange(g), m)) # the randomization
x = np.round(rng.normal(60, 10, n))             # pretest, measured before
    assignment
mu_true = np.array([70.0, 74.0, 71.0, 77.0])
y = np.round(mu_true[group] + 0.8 * (x - 60) + rng.normal(0, 6, n), 1)

Z = np.eye(g)[group]                         # indicator matrix of the
    formats
n_k = Z.sum(axis=0)
```

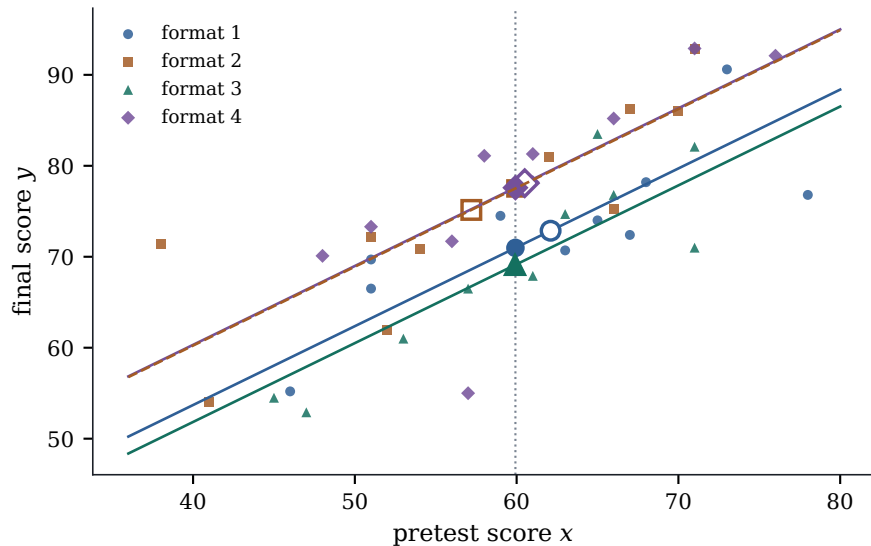


Figure 18.1.1: The tutoring experiment: final score against pretest score for the four formats, with the fitted parallel lines of the covariance model. Open symbols mark the raw group means $(\bar{x}_k, \bar{y}_k)$, and filled symbols on the dotted vertical line mark the adjusted means at the overall pretest mean $\bar{x}$. The lines of formats 2 (dashed, drawn on top) and 4 almost coincide.

```
x_bar, y_bar = Z.T @ x / n_k, Z.T @ y / n_k          # group means

def within(v):
    """(I - M) v: deviations from the group means (M projects onto C(Z))."""
    return v - (Z.T @ v / n_k)[group]

E_xx, E_xy, E_yy = within(x) @ within(x), within(x) @ within(y), within(y) @ within(
    y)
beta = E_xy / E_xx                                  # slope from within-group
            variation only
sse = E_yy - E_xy ** 2 / E_xx                       # ANCOVA residual sum of
            squares
nu = n - g - 1

T_xx = np.sum((x - x.mean()) ** 2)                  # the same with I - MO, MO =
            mean only
T_xy = np.sum((x - x.mean()) * (y - y.mean()))
T_yy = np.sum((y - y.mean()) ** 2)
sse0 = T_yy - T_xy ** 2 / T_xx                      # reduced model: one line for
            everyone
ss_trt = sse0 - sse                                # adjusted treatment sum of
            squares
F_adj = (ss_trt / (g - 1)) / (sse / nu)
F_unadj = ((T_yy - E_yy) / (g - 1)) / (E_yy / (n - g))
print(f"slope {beta:.4f}; SSE: ANOVA {E_yy:.2f} on {n - g} df, ANCOVA {sse:.2f} on
      {nu} df")
print(f"treatments: unadjusted F = {F_unadj:.2f} (p = {stats.f.sf(F_unadj, g - 1, n
      - g):.3f}),"
      f" adjusted F = {F_adj:.2f} (p = {stats.f.sf(F_adj, g - 1, nu):.4f})")
```

[Run this code online](#)

18.1.4 What a covariate does in a randomized experiment

In a randomized experiment the groups differ on the covariate only by chance. Adjustment still helps, for two reasons.

Precision. The error variance of the covariance model is the variance of y given x , which is smaller whenever the two are correlated. The price is the estimated slope, which adds the second term in Eq. 18.1.6. The next result balances gain against price when the covariate is random, as it is in practice.

Proposition 18.1.2 — The precision of adjustment

Consider a one-way layout with g groups of m units, $N = gm$. Suppose the covariate is measured before randomization, so that the x_i are independent $N(\xi, \tau^2)$ whatever their group, and that given all the x 's the y_i are independent and normal, with $E(y_i | x_i) = \mu_k + \beta x_i$ for unit i in group k and $\text{Var}(y_i | x_i) = \sigma^2$. Let ρ be the within-group correlation of x and y . For two groups $k \neq l$, the unadjusted estimate $\bar{y}_k - \bar{y}_l$ and the adjusted estimate $(\bar{y}_k - \bar{y}_l) - \hat{\beta}(\bar{x}_k - \bar{x}_l)$ are both unbiased for $\mu_k - \mu_l$. If $N - g > 2$,

$$\frac{\text{Var}(\text{adjusted})}{\text{Var}(\text{unadjusted})} = (1 - \rho^2) \left(1 + \frac{1}{N - g - 2} \right). \quad (18.1.8)$$

Proof. Write $\bar{y}_k = \mu_k + \beta \bar{x}_k + \bar{\varepsilon}_k$, where $\varepsilon_i = y_i - \mu_k - \beta x_i$ has mean zero and variance σ^2 and is independent of all the x 's. Unconditionally $\bar{y}_k - \bar{y}_l$ has mean $\mu_k - \mu_l$, because $E \bar{x}_k = E \bar{x}_l$, and variance $(2/m)(\beta^2 \tau^2 + \sigma^2)$. The within-group variance of y is $\beta^2 \tau^2 + \sigma^2$, and $\sigma^2 = (1 - \rho^2)(\beta^2 \tau^2 + \sigma^2)$.

Given all the x 's, the covariance model holds with fixed covariates. By Theorem 18.1.1(e), with $\mathbf{p}$ the difference of the two scaled group indicators (so $\mathbf{M}\mathbf{p} = \mathbf{p}$ and $\mathbf{p}^\top \mathbf{Z} = \bar{x}_k - \bar{x}_l$), the adjusted estimate is conditionally unbiased with conditional variance $\sigma^2 \{2/m + (\bar{x}_k - \bar{x}_l)^2 / E_{xx}\}$. Its unconditional mean is therefore $\mu_k - \mu_l$, and its unconditional variance is the mean of the conditional variance. The group means of x are independent of the within-group sum of squares E_{xx} (Corollary 4.4.5, applied group by group), with $\bar{x}_k - \bar{x}_l \sim N(0, 2\tau^2/m)$ and $E_{xx}/\tau^2 \sim \chi^2(N - g)$. For $V \sim \chi^2(\nu)$ with $\nu > 2$, $E(1/V) = 1/(\nu - 2)$ (Exercise (B1)). Hence

$$E \frac{(\bar{x}_k - \bar{x}_l)^2}{E_{xx}} = \frac{2\tau^2}{m} \cdot \frac{1}{\tau^2(N - g - 2)}, \quad \text{Var}(\text{adjusted}) = \frac{2\sigma^2}{m} \left(1 + \frac{1}{N - g - 2} \right).$$

Dividing by the unadjusted variance gives Eq. 18.1.8. ■

The factor $1 - \rho^2$ is what adjustment would achieve if the slope were known. The factor $1 + 1/(N - g - 2)$ is the cost of estimating it. For the design of Example (Four study formats with a pretest) the cost is 1.0294. Adjustment pays whenever $\rho^2 > 1/(N - g - 1)$, that is, when $|\rho|$ exceeds 0.169. At $\rho = 0.8$, the correlation used to generate the tutoring data, the adjusted comparison has 0.371 of the variance of the unadjusted one (simulation: 0.372; Figure 18.1.2). Without the covariate the experiment would need almost three times as many students.

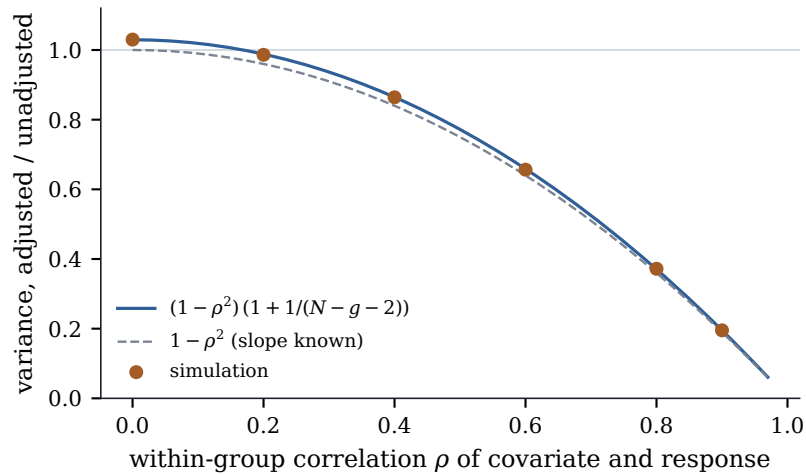


Figure 18.1.2: Variance of the adjusted difference of two group means relative to the unadjusted difference, for four groups of ten, as a function of the within-group correlation between covariate and response. The curve is Eq. 18.1.8, and the points are simulated from 100000 experiments each. The dashed curve is the ratio when the slope is known.

Chance imbalance. Randomization makes $\bar{y}_k - \bar{y}_l$ unbiased *over repetitions of the experiment*. In the experiment actually run, the proof above shows that, given the covariates,

$$E(\bar{y}_k - \bar{y}_l \mid x\text{'s}) = \mu_k - \mu_l + \beta(\bar{x}_k - \bar{x}_l),$$

which is *not* $\mu_k - \mu_l$ unless the covariate means agree. Once the pretests are known, the raw difference has an error of predictable sign. In [Example \(Four study formats with a pretest\)](#), format 1 started 4.9 points ahead of format 2 on the pretest, which predicts $0.8670 \times 4.9 \approx 4.25$ points of the final difference. The raw difference between formats 1 and 2 is -2.29 , and the conditionally unbiased adjusted difference is $-2.29 - 4.25 = -6.54$.

All this assumes the covariate was chosen in advance. Choosing, after seeing the data, the covariates that make the effect look best is a form of multiple testing, which is why trial protocols name their covariates beforehand.

Remark (Random covariates). The theorem ([Theorem 18.1.1](#)) treats $\mathbf{Z}$ as fixed. If, given random covariates, the model holds with normal errors, every F test and t interval of the theorem keeps its level unconditionally (Theorem 14.1.2, applied after reparameterizing the design part to full rank, which changes neither the tests nor the estimable functions). Only the power depends on the covariate values that occur.

18.1.5 Adjusting for a covariate that the treatment affects

Warning. A covariate must not be affected by the treatment. If it is measured after assignment and the treatment changes it, adjusting for it removes the part of the treatment effect that acts through the covariate, and the adjusted comparison no longer estimates the effect of assigning the treatment.

If hours of study were used as a covariate, a format that works *by* getting students to study more would be compared with the others at equal hours of study, which removes the very thing

it does. In a synthetic randomized experiment with two arms of 100, let the treatment raise a post-assignment variable w by 2 units and let $E(y) = \theta T + 0.75w$ with direct effect $\theta = 1$. The effect of assigning the treatment is $1 + 0.75 \times 2 = 2.50$. Over 20000 replications, the unadjusted difference of means averages 2.500, and the coefficient of the treatment after adjustment for w averages 1.000, the direct effect, which is not what a decision about the treatment needs.

```
import numpy as np

rng = np.random.default_rng(2025)
n, reps = 200, 20_000
theta, delta, gamma = 1.0, 2.0, 0.75          # direct effect; T -> w; w ->
y                                              # effect of assigning
total = theta + gamma * delta
treatment
est_unadj, est_adj = np.empty(reps), np.empty(reps)
for r in range(reps):
    T = rng.permutation(np.repeat([0.0, 1.0], n // 2)) # randomized assignment
    w = rng.normal(size=n) + delta * T                 # measured after treatment
    y = theta * T + gamma * w + rng.normal(size=n)
    X = np.column_stack([np.ones(n), T, w])
    est_unadj[r] = y[T == 1].mean() - y[T == 0].mean()
    est_adj[r] = np.linalg.lstsq(X, y, rcond=None)[0][1]
print(f"total effect {total:.2f}: unadjusted mean {est_unadj.mean():.3f},"
      f" adjusted for w mean {est_adj.mean():.3f}")
```

[Run this code online](#)

Chapter 25 develops the language for this in observational data: mediators, on the causal path from treatment to response, must not be adjusted for when the total effect is wanted. In a randomized experiment the safe rule is simple: *adjust only for variables fixed before randomization*. By Proposition 18.1.2 this costs little even when the covariate turns out to be useless. A test of whether the groups are “balanced” on such a covariate is pointless, since its null hypothesis is true by construction. What matters is whether the covariate predicts the response.

18.1.6 Exercises

18.1.6.1 A. Check your understanding

Exercise A1.

In the one-way layout with one covariate, suppose x is constant within each group. Show that Eq. 18.1.3 fails, and describe the set of least squares estimates of β . Is the adjusted difference of two group means estimable?

Solution. If $x_i = c_k$ for every unit in group k , then $\mathbf{x}$ is a combination of the group indicators, so $\mathbf{x} \in \mathcal{C}(\mathbf{X})$, $(\mathbf{I} - \mathbf{M})\mathbf{x} = \mathbf{0}$, and Eq. 18.1.3 fails. The mean of group k is $\theta_k = \mu_k + \beta c_k$, and only the θ_k are estimable. Every value of β belongs to some least squares estimate: for any b , the group parameters $\bar{y}_k - bc_k$ together with b reproduce the fitted group means. The adjusted difference $\mu_k - \mu_l = (\theta_k - \theta_l) - \beta(c_k - c_l)$ is therefore estimable iff $c_k = c_l$. The data cannot separate a covariate that varies only between groups from the group effects.

Exercise A2.

From the numbers in [Example \(Four study formats with a pretest\)](#), compute the F statistic for $\beta = 0$ and check that its square root is the t statistic $\hat{\beta}/(s/\sqrt{E_{xx}})$.

18.1.6.2 B. Practice**Exercise B1.**

Let $V \sim \chi^2(\nu)$ with $\nu > 2$. Show that $E(1/V) = 1/(\nu - 2)$, and that $E(1/V) = \infty$ for $\nu \leq 2$.

Solution. The density of V is $f_\nu(v) = v^{\nu/2-1}e^{-v/2}/\{2^{\nu/2}\Gamma(\nu/2)\}$. Then

$$E(1/V) = \int_0^\infty \frac{v^{\nu/2-2}e^{-v/2}}{2^{\nu/2}\Gamma(\nu/2)} dv = \frac{2^{\nu/2-1}\Gamma(\nu/2-1)}{2^{\nu/2}\Gamma(\nu/2)} = \frac{1}{2(\nu/2-1)} = \frac{1}{\nu-2},$$

using $\Gamma(s+1) = s\Gamma(s)$. The integral converges at 0 iff $\nu/2 - 2 > -1$, that is, iff $\nu > 2$.

Exercise B2.

In the one-way layout with one covariate, write $B_{xx} = T_{xx} - E_{xx}$, and similarly B_{xy} and B_{yy} , for the between-groups sums of squares and products. Show that the adjusted treatment sum of squares equals

$$B_{yy} - \left(\frac{T_{xy}^2}{T_{xx}} - \frac{E_{xy}^2}{E_{xx}} \right),$$

and deduce that it exceeds B_{yy} iff $E_{xy}^2/E_{xx} > T_{xy}^2/T_{xx}$. Show that this happens in [Example \(Four study formats with a pretest\)](#), and explain it in terms of B_{xy} .

Solution. The formula is [Eq. 18.1.7](#) with $T_{yy} - E_{yy} = B_{yy}$. In the example $E_{xy}^2/E_{xx} = 2738.94$ and $T_{xy}^2/T_{xx} = 2595.83$, so the adjusted sum of squares is larger, by 143.11. The total regression uses both the within-group and the between-group variation of x . Here the between-group cross-product $B_{xy} = -31.4$ is negative: across the four formats, higher pretest means go with *lower* final means, the opposite of the within-group relationship. The single line for all students therefore explains less than the within-group lines do, and the difference is attributed to the formats.

Exercise B3.

In the one-way layout with q covariates, use [Eq. 18.1.6](#) to show that the adjusted difference $(\bar{y}_k - \bar{y}_l) - (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \hat{\boldsymbol{\gamma}}$ has variance $\sigma^2\{1/n_k + 1/n_l + (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)\}$.

18.1.6.3 C. Going deeper**Exercise C1.**

In the one-way covariance model, show that replacing x by $a + cx$ with $c \neq 0$ leaves the

fitted values, SSE, the adjusted treatment sum of squares and the adjusted differences unchanged. Then replace x by $x + d_k$ on group k , with group-specific constants d_k . Show that $\hat{\beta}$ and SSE are unchanged but the adjusted differences and the adjusted treatment sum of squares change. Explain why a covariate recorded with a group-specific calibration error is dangerous.